Logistic Regression: Complete Formula Term Reference

Overview

This document summarizes all the key formulas used in Logistic Regression, along with explanations for each symbol, term name, and their role in the model. It is structured to help with manual implementation and understanding.

Step-by-Step Summary

Step 1: Linear Combination (Logit) Computes the raw score using feature values and learned parameters.

$$z = w^T x + b$$

Symbol	Symbol Name	Term Name	What It Does
z	z	Logit / Linear Score	Raw score before activation (input to sigmoid)
w	w	Weight Vector	Parameters that assign importance to features
x	x	Feature Vector	Input data
b	b	Bias / Intercept	Adjusts decision boundary
$w^T x$	w transpose x	Dot Product	Weighted sum of features

Step 2: Sigmoid Function Converts raw score z into a probability between 0 and 1.

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Symbol	Symbol Name	Term Name	What It Does
$\hat{y}$	y-hat	Predicted Probability	Model output: probability of class 1
$\sigma(z)$	sigma of z	Sigmoid Function	Maps raw score to probability space
z	z	Logit	Input to sigmoid function
e	e	Euler's Number	Constant for natural exponentiation

Step 3: Loss Function (Binary Cross-Entropy) Measures error between predicted probability and true label.

$$\mathcal{L}(\hat{y}, y) = -\left[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})\right]$$

Symbol	Symbol Name	Term Name	What It Does
$\mathcal{L}$	L	Loss Function	Quantifies error for one prediction
y	y	True Label	Actual class label (0 or 1)
$\hat{y}$	y-hat	Predicted Probability	Model output
$\log(\hat{y})$	log y-hat	Log-Likelihood	Penalizes incorrect predictions

Step 4: Cost Function (Mean Loss) Averages loss over the entire dataset.

$$J(w,b) = \frac{1}{m}\sum_{i=1}^{m}\mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$

Symbol	Symbol Name	Term Name	What It Does
$J(w,b)$	J	Cost Function	Measures total model error
m	m	Number of Samples	Size of dataset
$\sum$	summation	Total Loss	Adds up individual losses
$\mathcal{L}(\hat{y}^{(i)}, y^{(i)})$	Loss i	Per-sample Loss	Loss for each example

Step 5: Gradient Computation Computes direction to update weights and bias to reduce loss.

$$\frac{\partial J}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}(\hat{y}^{(i)} - y^{(i)})x^{(i)} \qquad \frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m}(\hat{y}^{(i)} - y^{(i)})$$

Symbol	Symbol Name	Term Name	What It Does
$\frac{\partial J}{\partial w}$	dJ/dw	Gradient (Weights)	Shows how to change each weight
$\frac{\partial J}{\partial b}$	dJ/db	Gradient (Bias)	Shows how to change bias
$(\hat{y}^{(i)} - y^{(i)})$	prediction error	Residual	Difference between prediction and truth
$x^{(i)}$	x-i	Feature Vector	Input data for sample i

Step 6: Gradient Descent Update Rule Updates parameters using gradients and learning rate.

$$w := w - \alpha \cdot \frac{\partial J}{\partial w} \qquad b := b - \alpha \cdot \frac{\partial J}{\partial b}$$

Symbol	Symbol Name	Term Name	What It Does
w, b	parameters	Weights and Bias	Model parameters to optimize
α	alpha	Learning Rate	Controls update size per step
$:=$	assignment	Update Operator	Applies parameter update
$\frac{\partial J}{\partial w}, \frac{\partial J}{\partial b}$	gradients	Slopes	Direction of steepest descent

Step 7: Final Prediction Rule Converts probability into binary class label.

$$\text{predict} = \begin{cases} 1 & \text{if } \hat{y} \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Symbol	Symbol Name	Term Name	What It Does
$\hat{y}$	y-hat	Predicted Probability	Output probability from model
Threshold	threshold	Decision Cutoff	Default is 0.5 for binary classification
0 or 1	class label	Final Prediction	Assigned class based on threshold